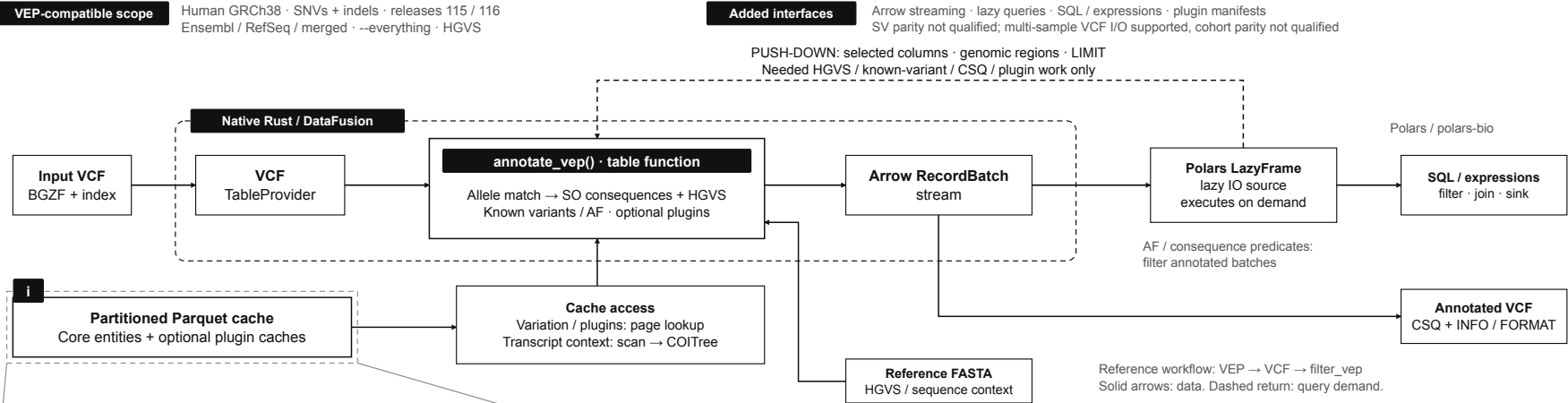
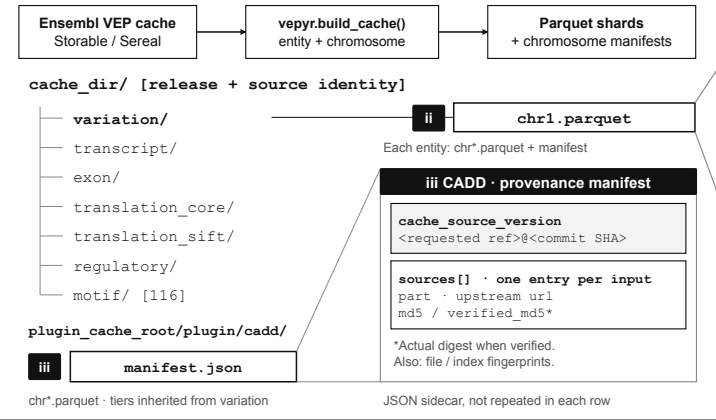


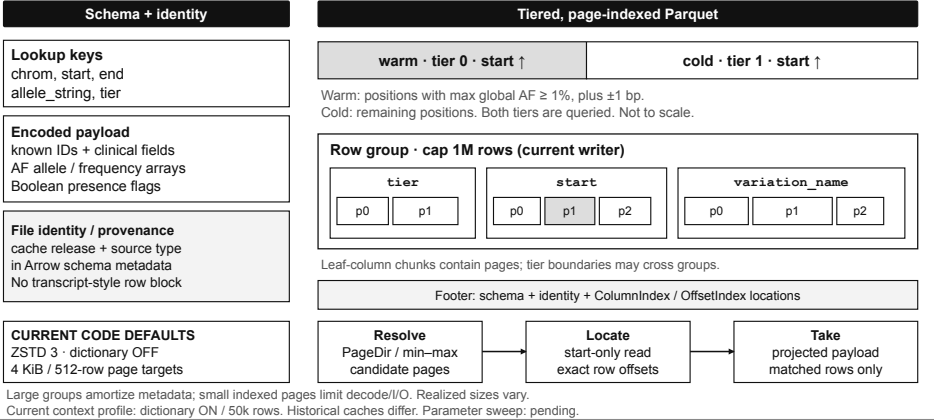
A1 Annotation architecture



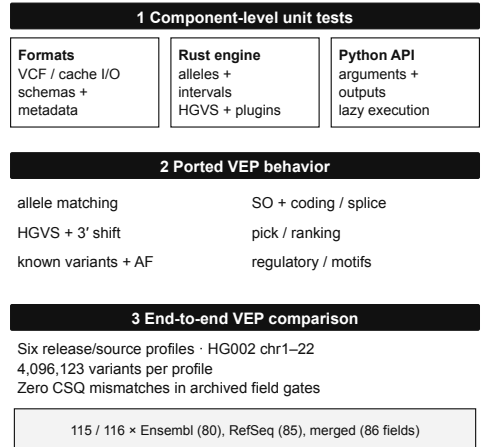
A2 Cache directory



A3 variation/chr1.parquet

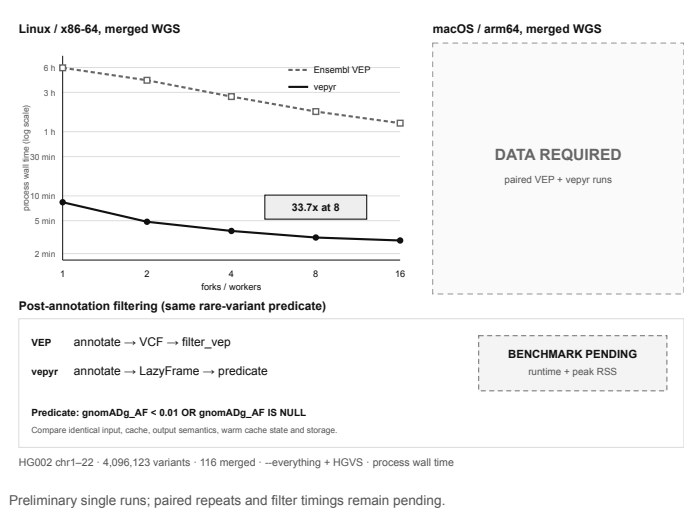


B Quality validation

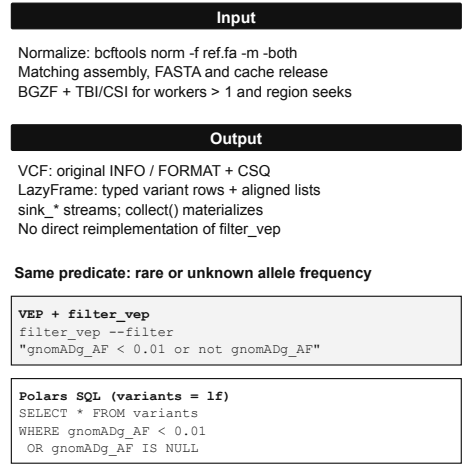


Separate harness: strict / canonical VCF-body MD5.
Field gates are not presented as MD5 validation.

C Performance



D Contracts + filtering



Per-CSQ predicates need aligned explode / any.
Ontology expansion must be explicit.